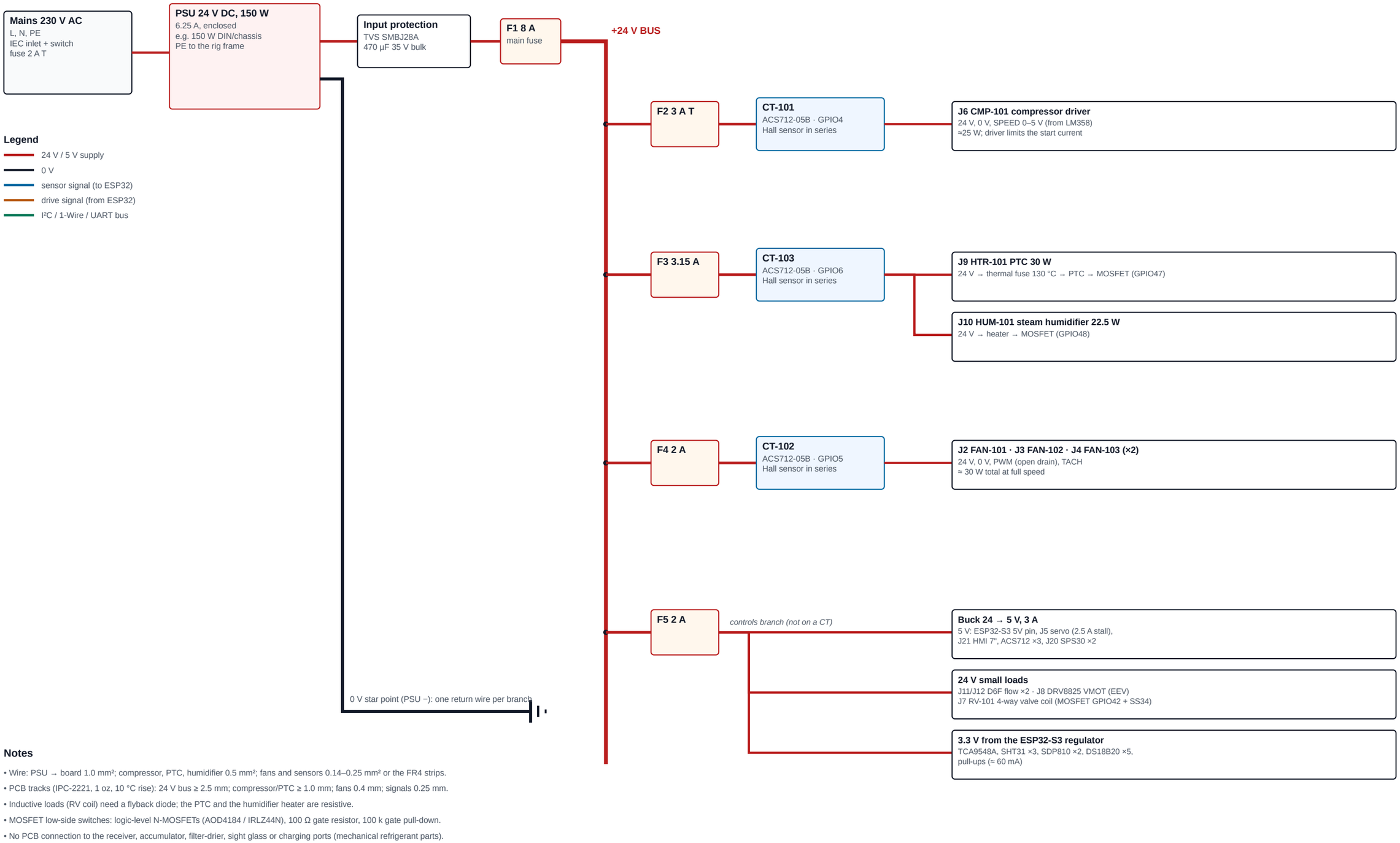
POWER DISTRIBUTION

All loads run from one 24 V DC supply. Each branch has its own fuse; the three ACS712 current sensors measure the compressor, fan and heater branches (CT-101 … 103). The controls branch is not on a CT (≈3.5 W, added in firmware).



Notes

- Wire: PSU → board 1.0 mm²; compressor, PTC, humidifier 0.5 mm²; fans and sensors 0.14–0.25 mm² or the FR4 strips.
- PCB tracks (IPC-2221, 1 oz, 10 °C rise): 24 V bus ≥ 2.5 mm; compressor/PTC ≥ 1.0 mm; fans 0.4 mm; signals 0.25 mm.
- Inductive loads (RV coil) need a flyback diode; the PTC and the humidifier heater are resistive.
- MOSFET low-side switches: logic-level N-MOSFETs (AOD4184 / IRLZ44N), 100 Ω gate resistor, 100 k gate pull-down.
- No PCB connection to the receiver, accumulator, filter-drier, sight glass or charging ports (mechanical refrigerant parts).

Clean-room heat-recovery air-pump rig

Sheet 1 · Power distribution, fusing and current sensing

ESP32-S3 controller · 24 V DC · pin map = firmware/include/config.h

CONTROLLER I/O

ESP32-S3-DevKitC-1 on the main PCB (back wall). Inputs on the left, outputs on the right. Every device is reached by its own FR4 strip with the connector shown (J-number). Interface parts sit on the main PCB.

SHT31 0x44 · TT-101 / MT-101 fresh air — J15 — ch0
SHT31 0x44 · TT-105 / MT-105 supply — J16 — ch1
SDP810-125Pa 0x25 · PDT-101 room ΔP — J17 — ch2
SDP810-500Pa 0x25 · PDT-102 HEPA ΔP — J18 — ch3
SHT31 0x44 · TT-106 / MT-106 room — J19 — ch4
SPS30 0x69 (5 V) · PT-101 room particles — J20 — ch5
SPS30 0x69 (5 V) · PT-101A fill-point particles — J20 — ch6

TCA9548A
I²C mux 0x70
2.2 k pull-ups on SDA/SCL and on every channel (3.3 V)

Each channel: 3.3 V (5 V for SPS30), 0 V, SDAx, SCLx. SPS30 SEL pin → 0 V (I²C mode).

TT-102 mix, TT-103 pre-heat, TT-104 coil · DS18B20 ×3 — J13 — 4.7 k to 3.3 V
TT-107 return, TT-108 exhaust · DS18B20 ×2 — J14 — 4.7 k to 3.3 V
FT-101 fresh-air flow · Omron D6F-W04A1 (24 V supply) — J11 — direct 0.5–2.5 V
FT-102 supply flow · Omron D6F-W04A1 (24 V supply) — J12 — direct 0.5–2.5 V
CT-101 compressor current · ACS712-05B (5 V) — PCB — 10 k / 20 k
CT-102 fan current · ACS712-05B (5 V) — PCB — 10 k / 20 k
CT-103 PTC + humidifier current · ACS712-05B — PCB — 10 k / 20 k
VT-101 24 V bus voltage — PCB — 100 k / 10 k
FAN-101 tach (open collector, 2 pulses/rev) — J2.4 — 10 k to 3.3 V
FAN-102 tach — J3.4 — 10 k to 3.3 V
FAN-103 tach (fan A) — J4.4 — 10 k to 3.3 V

5 V → servo, HMI, ACS712 ×3, SPS30 ×2
3.3 V → TCA9548A, SHT31 ×3, SDP810 ×2, DS18B20 ×5, pull-ups
0 V → common; ADC reference ground at the board

ESP32-S3-DevKitC-1

N8R8 · 3.3 V logic · ADC1 only

GPIO8	SDA			
GPIO9	SCL			
GPIO15	1-Wire A			
GPIO16	1-Wire B			
GPIO1	ADC1_CH0			
GPIO2	ADC1_CH1			
GPIO4	ADC1_CH3			
GPIO5	ADC1_CH4			
GPIO6	ADC1_CH5			
GPIO7	ADC1_CH6			
GPIO10	tach			
GPIO11	tach			
GPIO12	tach			
5V	from buck 24→5 V			
3V3	out, ≤ 300 mA			
GND	0 V star			

		PWM 25 kHz	GPIO38	AO3400 open drain — FAN-101 fresh-air compressor (24 V blower, PWM in) — J2
		PWM 25 kHz	GPIO39	AO3400 open drain — FAN-102 supply fan — J3
		PWM 25 kHz	GPIO40	AO3400 open drain — FAN-103 exhaust fans ×2 (PWM in parallel) — J4
		50 Hz servo	GPIO21	330 Ω series — SV-101 mixing-damper servo MG996R (5 V) — J5
		PWM 20 kHz	GPIO41	RC 10 k/1 µF → LM358 ×1.5 — CMP-101 compressor driver SPEED 0–5 V — J6
		digital	GPIO42	MOSFET + SS34 flyback — RV-101 4-way valve coil 24 V (high = cooling) — J7
		STEP	GPIO13	DRV8825 (VMOT 24 V) — EEV-101 bipolar stepper: A1 A2 B1 B2 — J8
		DIR	GPIO14	DRV8825 EN→0V, SLP/RST→3V3 — (Vref = I_coil × 0.5 Ω)
		PWM 1 kHz	GPIO47	AOD4184 low side — HTR-101 PTC reheat 30 W + 130 °C thermal fuse — J9
		PWM 1 kHz	GPIO48	AOD4184 low side — HUM-101 steam humidifier heater 22.5 W — J10
		UART1 TX	GPIO17	direct (3.3 V) — HMI-101 RX · 7" serial HMI, 115200 8N1 (5 V) — J21
		UART1 RX	GPIO18	1 k series — HMI-101 TX — J21

USB-C (native USB): flashing + JSON telemetry to the laptop (Live monitor).

Spare: GPIO3/45/46 are strapping pins; GPIO35–37 used by octal PSRAM; leave free.

Compressor speed: 0–3.3 V PWM → RC (fc 16 Hz) → LM358 gain 1 + 5.1k/10k = 1.51 → 0–5 V.

Fans: open-drain inverts PWM (firmware FAN_PWM_INVERTED = true).

Legend

- 24 V / 5 V supply
- 0 V
- sensor signal (to ESP32)
- drive signal (from ESP32)
- I²C / 1-Wire / UART bus

Clean-room heat-recovery air-pump rig

Sheet 2 · Controller I/O: ESP32-S3 pins, interfaces, devices

ESP32-S3 controller · 24 V DC · pin map = firmware/include/config.h